Supplementary Material

Which reverts count? Successor filtering and observation windows in Wikipedia revert-based metrics

All tables are generated by 03_pipeline/12_supplement.py from the released aggregates in 04_output/. Editions are ordered by the bot-and-tool share of first reverts. Confirmatory editions: jawiki, plwiki, itwiki.

Table S1. Filter effect by observation window (replace variant, F2), R(W) and substitution S(W)

Δ in percentage points from the 60,000-anchor samples (all anchors in kowiki); R(W) = Δ(W)/Δ(24 h); S(W) = 1 − Δ(replace, W)/Δ(strict, W) on the same sample.

Δ(F2 replace, W), pp

	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
12 h	18.65	15.70	9.70	9.52	6.12	6.30	3.00
24 h	18.83	15.90	9.83	9.59	6.22	6.36	2.97
48 h	18.79	15.79	9.76	9.56	6.24	6.34	2.94
7 d	18.46	15.17	9.43	9.48	6.11	6.17	2.94
30 d	17.87	14.33	8.84	9.31	5.89	5.89	2.87
180 d	17.44	13.62	7.95	8.92	5.63	5.56	2.75
unbounded	17.37	13.41	7.62	8.79	5.45	5.45	2.73

R(W)

	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
12 h	0.990	0.987	0.987	0.993	0.984	0.991	1.008
24 h	1.000	1.000	1.000	1.000	1.000	1.000	1.000
48 h	0.998	0.993	0.993	0.997	1.004	0.997	0.990
7 d	0.980	0.954	0.959	0.989	0.983	0.970	0.988
30 d	0.949	0.901	0.899	0.971	0.947	0.926	0.965
180 d	0.926	0.857	0.809	0.931	0.905	0.875	0.926
unbounded	0.922	0.843	0.775	0.917	0.877	0.858	0.919

S(W)

	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
12 h	0.087	0.102	0.100	0.079	0.098	0.109	0.131
24 h	0.100	0.114	0.122	0.089	0.111	0.120	0.145
48 h	0.116	0.130	0.150	0.099	0.122	0.133	0.161
7 d	0.147	0.174	0.210	0.117	0.154	0.168	0.180
30 d	0.186	0.227	0.278	0.145	0.205	0.216	0.211
180 d	0.212	0.267	0.356	0.186	0.256	0.264	0.251
unbounded	0.218	0.279	0.385	0.198	0.280	0.279	0.258

Table S2. Strict filter effect by definition of the excluded class (24-hour window, all primary anchors)

Δ = p(F0) − p(F) in percentage points, 95 percent analytic interval in Table S2 data file. F1h/F1c: bot flag from the historical / current column only; F1fast and F2fast additionally treat registered non-bot reverts within 5 seconds as bots; F2s adds mw-undo to the tool tags; F3 additionally excludes anonymous reverters.

variant	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
F1 bots (current or historical)	4.53	0.01	0.07	0.04	0.03	1.02	0.01
F1c bots (current only)	4.53	0.01	0.03	0.04	0.02	1.02	0.01
F1fast bots + <=5 s registered	4.57	0.02	0.08	0.07	0.03	1.03	0.01
F1h bots (historical only)	4.53	0.01	0.07	0.04	0.03	1.02	0.01
F2 bots + tools (rollback/huggle/twinkle)	21.09	17.94	11.19	10.54	7.01	7.22	3.46
F2fast bots + tools + <=5 s	21.10	17.94	11.20	10.54	7.01	7.23	3.46
F2s bots + tools + undo	32.33	28.11	17.06	17.85	14.56	18.91	9.15
F3 bots + tools + anonymous reverters	21.92	18.77	13.27	10.96	7.68	7.53	4.57

Unbounded window:

variant	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
F1 bots (current or historical)	4.58	0.03	0.16	0.08	0.06	1.06	0.06
F1c bots (current only)	4.58	0.03	0.08	0.08	0.04	1.06	0.06
F1fast bots + <=5 s registered	4.63	0.05	0.18	0.12	0.06	1.06	0.06
F1h bots (historical only)	4.58	0.03	0.16	0.08	0.06	1.06	0.06
F2 bots + tools (rollback/huggle/twinkle)	22.38	18.58	12.38	10.94	7.57	7.55	3.65
F2fast bots + tools + <=5 s	22.39	18.58	12.39	10.94	7.57	7.55	3.65
F2s bots + tools + undo	34.97	29.41	19.83	19.50	18.95	20.34	10.52
F3 bots + tools + anonymous reverters	23.77	19.83	16.36	11.45	8.51	8.16	5.53

Table S3. Cohort variants: relative strict filter effect Δ(F2, 24 h)/p(F0, 24 h) and cohort size

cohort	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
all registered first edits 2019+	0.565	0.523	0.511	0.398	0.317	0.292	0.259
self-created first edits 2019+ (primary)	0.570	0.525	0.519	0.411	0.316	0.297	0.248
self-created, anchors 2021+ (all tags available)	0.649	0.546	0.555	0.395	0.305	0.282	0.301
self-created, secondary anchors (edits 2-5 within 91 d)	0.564	0.517	0.512	0.409	0.335	0.315	0.280

Anchors per cohort:

cohort	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
all registered first edits 2019+	2488692	168749	59618	289854	85443	413873	247119
self-created first edits 2019+ (primary)	2189284	135400	47633	236903	67755	346938	223045
self-created, anchors 2021+ (all tags available)	1453656	88005	32751	159216	45382	229556	148508
self-created, secondary	2947210	181835	84573	282617	87029	434787	403249

cohort	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
anchors (edits 2-5 within 91 d)							

Table S4. Normal approximations to the registered one-sided tests, with Holm adjustment within edition

The registered tests are bootstrap bounds (reported in the main text); the normal approximation uses the analytic standard error of the paired difference (H1), the delta method (H2) and the log odds-ratio variance (H4). H3 is a descriptive agreement criterion and is not a test.

wiki	test	estimate	se	p_one_sided	p_holm
enwiki	H1 D=Δ(24h)−Δ(∞) > 0	1.463	0.078	< .001	< .001
enwiki	H2 S < 0.25	0.100	0.003	< .001	< .001
enwiki	H4 OR_max/OR_min >= 1.5 (log ratio vs log 1.5)	1.785	0.007	< .001	< .001
itwiki	H1 D=Δ(24h)−Δ(∞) > 0	2.492	0.077	< .001	< .001
itwiki	H2 S < 0.25	0.114	0.003	< .001	< .001
itwiki	H4 OR_max/OR_min >= 1.5 (log ratio vs log 1.5)	1.783	0.028	< .001	< .001
kowiki	H1 D=Δ(24h)−Δ(∞) > 0	2.213	0.091	< .001	< .001
kowiki	H2 S < 0.25	0.122	0.004	< .001	< .001
kowiki	H4 OR_max/OR_min >= 1.5 (log ratio vs log 1.5)	3.008	0.051	< .001	< .001
dewiki	H1 D=Δ(24h)−Δ(∞) > 0	0.793	0.052	< .001	< .001
dewiki	H2 S < 0.25	0.089	0.004	< .001	< .001
dewiki	H4 OR_max/OR_min >= 1.5 (log ratio vs log 1.5)	1.629	0.021	< .001	< .001
plwiki	H1 D=Δ(24h)−Δ(∞) > 0	0.767	0.054	< .001	< .001
plwiki	H2 S < 0.25	0.111	0.005	< .001	< .001
plwiki	H4 OR_max/OR_min >= 1.5 (log ratio vs log 1.5)	1.425	0.038	0.912	0.912
frwiki	H1 D=Δ(24h)−Δ(∞) > 0	0.905	0.050	< .001	< .001
frwiki	H2 S < 0.25	0.120	0.005	< .001	< .001
frwiki	H4 OR_max/OR_min >= 1.5 (log ratio vs log 1.5)	1.512	0.018	0.333	0.333
jawiki	H1 D=Δ(24h)−Δ(∞) > 0	0.242	0.032	< .001	< .001
jawiki	H2 S < 0.25	0.145	0.008	< .001	< .001
jawiki	H4 OR_max/OR_min >= 1.5 (log ratio vs log 1.5)	1.836	0.024	< .001	< .001

Table S5. Logistic regression of reverted status on filter and window (replace sample, long format), editor-clustered standard errors

Reference cell F0 at 24 h. Rows: anchors × 3 filters × 3 windows (about 540,000 per edition; 428,661 in kowiki); clusters = editors. Coefficients on the log-odds scale; cluster-robust (CR1) standard errors in parentheses.

term	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
const	-0.544 (0.008)	-0.66 (0.009)	-1.29 (0.011)	-1.069 (0.009)	-1.259 (0.01)	-1.136 (0.01)	-1.813 (0.012)

term	enwiki	itwiki	kowiki	dewiki	plwiki	frwiki	jawiki
F1r	-0.132 (0.003)	-0.0 (0.0)	-0.003 (0.001)	-0.003 (0.0)	-0.002 (0.0)	-0.034 (0.002)	-0.001 (0.0)
F2r	-0.98 (0.009)	-0.845 (0.008)	-0.726 (0.01)	-0.591 (0.008)	-0.407 (0.006)	-0.384 (0.006)	-0.272 (0.006)
W7d	0.091 (0.003)	0.067 (0.002)	0.151 (0.004)	0.091 (0.003)	0.106 (0.003)	0.071 (0.003)	0.112 (0.004)
Winf	0.172 (0.003)	0.121 (0.003)	0.341 (0.006)	0.196 (0.004)	0.419 (0.006)	0.138 (0.003)	0.244 (0.005)
F1r_x_W7d	0.016 (0.001)	0.0 (0.0)	-0.0 (0.001)	-0.0 (0.0)	-0.0 (0.0)	0.004 (0.001)	-0.0 (0.0)
F1r_x_Winf	0.034 (0.002)	-0.0 (0.0)	-0.002 (0.001)	-0.002 (0.0)	-0.001 (0.0)	0.009 (0.001)	-0.002 (0.001)
F2r_x_W7d	0.072 (0.003)	0.078 (0.003)	0.116 (0.005)	0.042 (0.002)	0.036 (0.002)	0.029 (0.002)	0.027 (0.002)
F2r_x_Winf	0.168 (0.005)	0.2 (0.005)	0.307 (0.008)	0.119 (0.004)	0.132 (0.004)	0.086 (0.003)	0.067 (0.003)

Table S6. kowiki: anchor definition (first edit, registered) versus first main-namespace edit (pilot definition), strict F2

anchor	W	n	p_F0	p_F2	delta_pp	ci_lo	ci_hi	rel_delta
first edit (registered)	W24h	47633	0.216	0.104	11.194	10.911	11.477	0.519
first edit (registered)	W7d	47633	0.243	0.123	11.937	11.646	12.228	0.492
first edit (registered)	Winf	47633	0.279	0.155	12.382	12.086	12.678	0.444
first ns0 edit (pilot)	W24h	52819	0.218	0.105	11.267	10.997	11.536	0.517
first ns0 edit (pilot)	W7d	52819	0.244	0.124	11.984	11.707	12.261	0.491
first ns0 edit (pilot)	Winf	52819	0.279	0.155	12.412	12.131	12.693	0.444

Table S6b. kowiki with and without the June 2023 to December 2024 period of elevated newcomer inflow.

cohort	n	p_F0_W24h	strict_delta_W24h	rel_W24h	replace_delta_W24h	p_F0_Winf	strict_delta_Winf	rel_Winf	replace_delta_Winf	R
all	47633	0.216	11.194	0.519	9.830	0.279	12.382	0.444	7.617	0.775
excl. 2023-06..2024-12 surge	35065	0.165	7.227	0.437	6.394	0.226	8.199	0.363	5.533	0.865
surge only	12568	0.357	22.263	0.624	19.416	0.427	24.053	0.563	13.432	0.692

Table S7. Simulation: R = Δ(∞)/Δ(24 h) as a function of the human re-revert probability after an excluded fast revert and the median human delay

Grid over revert base rate (.10, .25, .40), excluded-actor share (0 to .8) and excluded-actor speed (5 to 60 s); R is averaged over those parameters (min and max shown) because they do not affect it. Observed R in the seven editions: .78 to .92.

p(human re-revert)	median human delay	mean	min	max
0.000	10 min	1.000	1.000	1.000
0.000	1 h	1.000	1.000	1.000

p(human re-revert)	median human delay	mean	min	max
0.000	1 d	1.000	1.000	1.000
0.000	7 d	1.000	1.000	1.000
0.100	10 min	1.000	1.000	1.000
0.100	1 h	1.000	0.999	1.000
0.100	1 d	0.948	0.947	0.951
0.100	7 d	0.905	0.901	0.908
0.200	10 min	1.000	1.000	1.000
0.200	1 h	0.999	0.999	0.999
0.200	1 d	0.889	0.884	0.893
0.200	7 d	0.808	0.804	0.813
0.500	10 min	1.000	1.000	1.000
0.500	1 h	0.996	0.995	0.997
0.500	1 d	0.667	0.663	0.672
0.500	7 d	0.513	0.507	0.518

Table S8. Change-tag availability on reverting revisions, by edition and year (all namespaces, all users)

Counts of reverting revisions carrying each tag. twinkle is tagged only in enwiki from September 2020; huggle tagging in enwiki is nearly absent in 2021 and 2022.

mw-rollback

wiki	2018	2019	2020	2021	2022	2023	2024	2025	2026
enwiki	1122426	895626	932095	883106	750250	774662	589221	606866	445436
itwiki	119804	124817	147566	159560	156810	127405	103429	90733	44813
kowiki	20961	14881	28911	22528	31437	27802	29237	28939	11935
dewiki	193873	189579	154538	142501	117173	120366	97322	69006	44425
plwiki	33863	33984	42665	37281	36801	33087	34604	22962	12179
frwiki	82033	86959	68743	76641	67687	60521	59402	47767	33146
jawiki	9121	11759	18030	27752	24273	26174	19051	14461	8951

huggle

wiki	2018	2019	2020	2021	2022	2023	2024	2025	2026
enwiki	345176	262033	162423	1631	2409	151939	132999	104257	47488
itwiki	0	0	0	0	0	0	0	0	0
kowiki	0	0	0	0	0	0	0	0	0
dewiki	0	0	0	0	0	0	0	0	0
plwiki	0	0	0	0	0	0	0	0	0
frwiki	5016	4433	836	1111	1774	4027	1868	1510	280
jawiki	0	0	0	0	0	0	0	0	0

twinkle

wiki	2018	2019	2020	2021	2022	2023	2024	2025	2026
enwiki	0	0	207362	602187	543831	514898	541969	538128	322779
itwiki	0	0	0	0	0	0	0	0	0
kowiki	0	0	0	0	0	0	0	0	0
dewiki	0	0	0	0	0	0	0	0	0
plwiki	0	0	0	0	0	0	0	0	0
frwiki	0	0	0	0	0	0	0	0	0

wiki	2018	2019	2020	2021	2022	2023	2024	2025	2026
jawiki	0	0	0	0	0	0	0	0	0

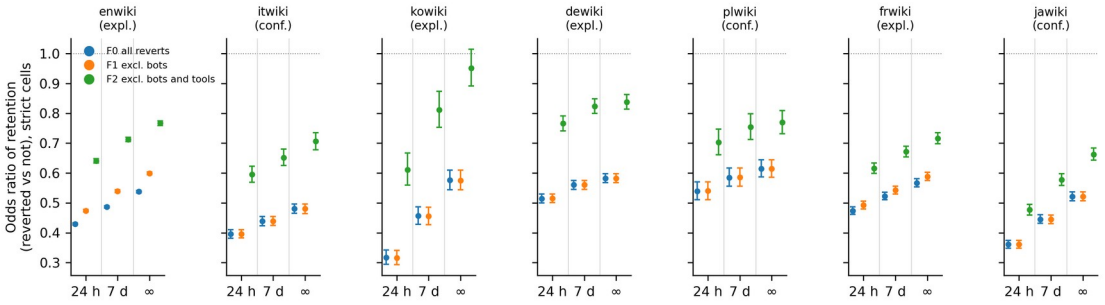
mw-undo

wiki	2018	2019	2020	2021	2022	2023	2024	2025	2026
enwiki	1494822	1529089	1540633	1552495	1435631	1340784	1362885	1285981	803483
itwiki	95578	117098	116429	114948	97846	89979	84370	73064	45823
kowiki	30946	29764	57931	45761	26982	21147	28007	21249	11311
dewiki	109547	96185	98655	97514	81234	79920	75943	69172	44466
plwiki	24450	26055	33152	46991	49128	31284	32403	22045	13004
frwiki	161866	154328	168055	151144	134478	137671	119977	118490	84144
jawiki	74816	95772	95019	93695	94968	67232	50025	44876	31462

mw-reverted

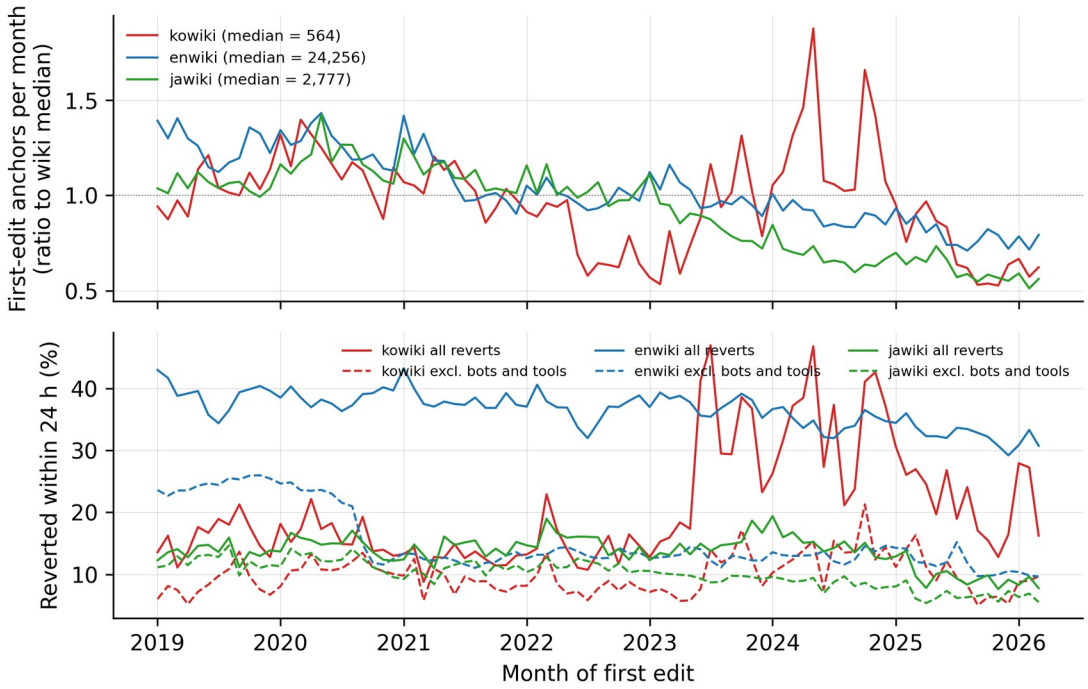
wiki	2018	2019	2020	2021	2022	2023	2024	2025	2026
enwiki	3780	5448	187989	592840	502091	472535	464239	453555	284397
itwiki	60	194	9389	30391	30547	25426	21420	19383	11200
kowiki	133	350	17909	42624	29790	23156	36357	33781	18214
dewiki	258	339	19656	65817	77713	56409	52028	50042	32436
plwiki	1070	936	6384	19635	17635	14595	20480	9618	5891
frwiki	323	501	10836	33358	26666	25423	24698	23414	16344
jawiki	101	236	13033	41124	42374	25019	21034	17799	11163

Figure S1. Retention odds ratio in the nine strict cells, by edition



Odds ratio of retention (an edit on days 2 to 60) for reverted versus non-reverted anchors, with 95 percent intervals, for the three filters (F0 all reverts, F1 excluding bots, F2 excluding bots and tools) in the 24-hour, 7-day and unbounded windows.

Figure S2. Monthly first-edit anchors and 24-hour revert rates, kowiki, enwiki and jawiki, 2019 to March 2026



Upper panel: anchors per month relative to each edition's median. Lower panel: share of anchors reverted within 24 hours under F0 (solid) and F2 strict (dashed). The kowiki inflow and revert-rate rise from mid-2023 to the end of 2024 is discussed in the Results; the drop in enwiki's F2 line in September 2020 marks the start of Twinkle tagging.